DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade)

Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru - 560 111. India

Department of Information Science and Engineering

(Accredited by NBA Tier 1: 2025-2028)

A Project Synopsis

on

Fovea: Zero-Instrumentation Payload Observability for Debugging and Root-Cause Analysis in Microservices deployed in Kubernetes ecosystems

Submitted as a part of the Final Year Project of

BACHELOR OF ENGINEERING

in

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

Submitted by

ABHAY SINGH	**1DS23IS005**
ADHISH SHUKLA	**1DS23IS008**
ADITYA	**1DS23IS012**
DHRUV PURI	**1DS23IS041**

Under the guidance of

Dr. Bindu Bhargavi SM
Assistant Professor
Dept. of ISE, DSCE

2026-2027

<table>
<tr><td>TITLE OF THE PROJECT</td><td>Fovea: Zero-Instrumentation Payload Observability for Debugging and Root-Cause Analysis in Microservices deployed in Kubernetes ecosystems</td></tr>
<tr><td>PROJECT GUIDE DETAILS</td><td>Dr. Bindu Bhargavi SM, Assistant Professor
PHONE NO.: +91 97314 25497
EMAIL ID: bindubhargavi-ise@dayanandasagar.edu</td></tr>
<tr><td>STUDENT DETAILS</td><td>
<table>
<tr><th>NAME</th><th>USN</th><th>PHONE NO.</th><th>EMAIL ID</th></tr>
<tr><td>Abhay Singh</td><td>1DS23IS005</td><td>9310511605</td><td>abhaysingh2115@gmail.com</td></tr>
<tr><td>Adhish Shukla</td><td>1DS23IS008</td><td>9111967315</td><td>adhishshukla12@gmail.com</td></tr>
<tr><td>Aditya</td><td>1DS23IS012</td><td>7204318987</td><td>pujeradi@gmail.com</td></tr>
<tr><td>Dhruv Puri</td><td>1DS23IS041</td><td>9149407326</td><td>dhruvpuri.35@gmail.com</td></tr>
</table>
</td></tr>
<tr><td>PROJECT TIMELINE (Tentative Start and End Date)</td><td>August 2026 - June 2027</td></tr>
<tr><td>PROJECT COORDINATORS</td><td>Dr. Santosh Anand, Prof. Ayush Mishra</td></tr>
<tr><td>DOMAIN OF THE PROJECT</td><td>Information Network Security and Privacy</td></tr>
<tr><td>BACKGROUND OF PROJECT WITH REGARD TO THE DRAWBACKS IN EXISTING PROJECTS</td><td>Modern applications are increasingly developed as microservices, where a single request may pass through multiple independent services. Understanding the complete flow of a request and identifying failures across these services is difficult without proper observability.</td></tr>
</table>

	Conventional approaches require developers to manually add tracing and instrumentation code to individual services, increasing development effort and application complexity. Existing eBPF-based observability solutions can automatically identify services and monitor network flows, but generally focus on metadata and do not provide visibility into the actual request and response payloads. This creates a gap in diagnosing application-level failures and understanding the actual data exchanged between services. Existing microservice observability solutions primarily rely on application-level instrumentation and distributed tracing frameworks, requiring modifications to application code. eBPF-based solutions such as Cilium and Hubble provide kernel-level visibility into network communication and service flows without modifying applications, but their observability is primarily focused on network and flow metadata rather than complete application payloads.
OBJECTIVES OF THE PROJECT	1. To develop a zero-instrumentation observability system for microservices using eBPF. 2. To capture and reconstruct HTTP request and JSON response payloads directly from kernel-level network communication. 3. To automatically identify and map communication flows between microservices. 4. To correlate captured payloads with their corresponding network flows and service calls. 5. To provide a unified web interface for viewing service maps, network flows, and exchanged payloads. 6. To deploy and evaluate the prototype in a Kubernetes-based microservice environment. 7. To measure payload reconstruction accuracy and evaluate the CPU and latency overhead introduced by the system.
PROBLEM STATEMENT	Existing microservice observability systems either require application-level instrumentation or provide only network-level metadata. As a result, developers may know that communication between services has failed but may not be able to inspect the actual request and response content responsible for the failure. There is a need for a zero-instrumentation observability system that can capture, reconstruct, and correlate both network flows and application payloads without modifying the applications being monitored.
SUMMARY OF THE PROJECT	Fovea is an end-to-end observability system designed for microservice environments. It uses eBPF to intercept data exchanged by services at the kernel level without requiring modifications to the monitored applications. An agent running on each machine in the cluster captures network data and reconstructs it into complete HTTP requests and JSON

	responses. The reconstructed payloads are correlated with their corresponding network flows and service calls. A web-based interface provides a unified view of the automatically generated service map, inter-service flows, and actual message contents. The prototype will be deployed on a Kubernetes-based microservice application and evaluated based on payload reconstruction accuracy, flow correlation, CPU overhead, and latency.
MODE OF CARRYING OUT THE PROJECT (Give details such as within the department/ external organization/ R&D centre details)	The project will be developed and evaluated within the Department of Information Science and Engineering, DSCE, Bangalore. The system will be implemented using eBPF (C/libbpf) for kernel-level data capture, Go for the per-node agent and backend collector, Cilium and Hubble for the underlying eBPF networking and flow layer, React for the web-based observability interface, and Kubernetes as the deployment and testing environment.
INTENDED BENEFICIARIES OF THE PROJECT	• Software developers working with microservice-based applications. • DevOps and Site Reliability Engineering (SRE) teams. • System and network administrators. • Cloud and Kubernetes administrators. • Organizations operating distributed and cloud-native applications. • Researchers and students working in network observability, eBPF, and distributed systems.
BASE PAPERS/ RELATED WORK	1. "eBPF-Based Network Monitoring Platform on Kubernetes" *2024 6th International Conference on Computer Communication and the Internet (ICCCI)*, IEEE, 2024, pp. 140–144. Author - Li-Der Chou, Luo-You Jian, and Yan-Wen Chen DOI: 10.1109/ICCCI62159.2024.10674074. 2. "Analyzing and Monitoring Kubernetes Microservices based on Distributed Tracing and Service Mesh" *2022 29th Asia-Pacific Software Engineering Conference (APSEC)*, IEEE, 2022. Author - Yu-Te Wang, Shang-Pin Ma, Yue-Jun Lai, and Yan-Cih Liang DOI: 10.1109/APSEC57359.2022.00066.
ABSTRACT	Fovea: eBPF based Zero-Instrumentation Payload Observability for Microservices addresses the challenge of understanding request behaviour in distributed microservice applications. Existing approaches often require developers to manually add tracing code to individual services, while existing eBPF-based observability tools primarily provide network-flow metadata without exposing the actual request and response content. Fovea proposes an end-to-end observability system

	that captures both network flows and application payloads without modifying the applications being observed. An agent running on each machine uses eBPF to intercept data as it passes through the kernel and reconstructs it into complete HTTP requests and JSON responses. The captured payloads are correlated with their corresponding network flows and service calls. A web interface provides a unified view of the automatically generated service map, inter-service flows, and actual message content. The prototype will be deployed on a Kubernetes-based microservice application and evaluated based on payload reconstruction accuracy, flow correlation, CPU overhead, and latency.
KEYWORDS	eBPF, Microservices, Observability, Zero Instrumentation, Payload Analysis, Network Security, Kubernetes, Cilium, Hubble, Distributed Systems

Signature of the Guide	**Signature of the Project Coordinators**